



UNDERSTANDING THE LABOUR MARKET OF HUMAN RESOURCES FOR HEALTH IN CAMEROON

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Abstract

Universal health coverage depends on having the necessary human resources to deliver health care services. Cameroon is among the African countries currently experiencing a crisis in the area of human resources for health (HRH). The major causes of the crisis include not only the poor production and recruitment planning of health personnel, but also shortcomings related to their management, which can be seen, for example, in the uneven distribution of existing health workers. Additional factors in the crisis include low salaries, poor working conditions and migration to developed countries. This document provides an overview of the HRH labour market in Cameroon, highlighting the importance of a comprehensive approach to understanding the driving forces that affect the supply and demand for health workers, in order to provide a basis for developing effective HRH policies that can contribute to progress towards universal health coverage.

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Acronyms and abbreviations used in this document

ALT	assistant laboratory technician
AN	assistant nurse
BUCREP	Central Bureau of Censuses and Population Studies
CALF	Cameroon Ad Lucem Foundation
CFAF	CFA franc
CHOC	Catholic Health Services Organization in Cameroon
CORDAID	Catholic Organization for Relief and Development Aid
CPCC	Council of Protestant Churches in Cameroon
C2D	Debt reduction and Development Contract
DH	district hospital
FMBS	Faculty of Medicine and Biomedical Sciences
GDP	gross domestic product
HF	health facility
HIPC	highly indebted poor countries
HRH	human resources for health
HSS	health sector strategy
HW	health worker
IHC	integrated health centre
MDG	Millennium Development Goal
MPA	minimum package of activities
NGO	nongovernmental organization
NA	nursing assistant
NIS	National Institute of Statistics
NSIF	National Security and Insurance Fund
PBF	performance-based financing
SMC	subdivisional medical centre
SRN	state registered nurse
WISN	Workload Indicators of Staffing Needs
WHO	World Health Organization

1. Introduction

Universal health coverage (UHC) seeks to ensure that: all people have access to the health services they need, whether promotive, preventive, curative, rehabilitative or palliative; that the services are of sufficient quality to be effective; and that the use of these services does not cause financial hardship (WHO, 2010). Universal health coverage depends on having the necessary human resources to deliver health care services. Human resources for health (HRH) include public and private sector doctors, nurses, midwives, pharmacists, technicians and other paraprofessional personnel, as well as untrained and informal-sector health workers, such as practitioners of traditional medicine, community health workers, and volunteers (WHO, 2006).

Cameroon is among the African countries currently experiencing a crisis in the area of human resources for health (HRH), which is adversely affecting progress towards the health-related Millennium Development Goals (MDGs). The major causes of the crisis include not only the poor production and recruitment planning of health personnel, but also shortcomings related to their management, which can be seen, for example, in the uneven distribution of existing health workers. In the public sector, there is a density of 0.09 medical doctors and 0.32 nurses per 1000 population at the national level. At the regional level, the Centre has a density of 0.27 medical doctors and 0.44 nurses per 1000 population, while the North has a 0.02 medical doctors and 0.19 nurses for every 1000 inhabitants (Ministry of Public Health, 2010). Additional factors in the crisis include low salaries, poor working conditions and migration to developed countries.

Some specific programmes, such as the emergency plan for upgrading quantitative and qualitative workforce (2006–2008) helped recruit nearly 2500 health workers under contract for the most deficient regions. These programmes were funded by development partners under the Contract Debt Relief and Development (C2D) and the Heavily Indebted Poor Countries (HIPC) Initiative. Once these came to an end, the staff were recruited into the public service, and were free to move to a different area.

From that perspective, the deficit in human resources becomes a question of managing and retaining existing staff in the so-called inaccessible areas. Although the country developed a health sector strategy (HSS) for 2001–2015, there was no specific policy for human resource development until the 2011–2015 National Development Plan for Health was issued in 2011. There was thus no specific HRH policy and management document, resulting in poor control of the workforce. Following a situation analysis in 2010, it was recommended that information on HRH be strengthened through the National HRH Observatory, in order to allow evidence-based policy development.

The present study assessed the national labour market in the health sector in Cameroon. The factors that determine supply and demand for health care were also examined and measured.

2. Country context

Cameroon is a central African country on the Gulf of Guinea, and had a population of 20 386 799 in 2012. The Cameroonian territory covers an area of 475 650 km², divided into ten regions with various regional population densities, which affect health service delivery and consequently the availability of human resources. The demographic loads vary from 4.1% in the South to 18.2% in the Centre. The population growth rate was estimated at 2.6% per year during 1987–2005. The population is mostly young: in 2010, 16.9% were under 5 years and 43.6% were under 15 years. Some 48% of the population live in rural areas (BUCREP, 2010).

With an annual growth rate greater than 3% since 2010 (4.6% in 2012), the gross domestic product (GDP) of Cameroon was estimated at 12 545.7 billion CFA francs (CFAF) (US\$ 24.47 billion)¹ in 2011 (INS, 2012a).

The poverty threshold in 2007 was 269 443 CFAF (US\$ 547) per year, or approximately US\$ 1.5 per day; 39.9% of the population lives on less than US\$ 1.5 per day. There are large differences between the regions; 9 out of 10 poor people live in rural areas. The poorest regions are the North and Far North, where the poverty rate is more than 60%, while in Yaoundé and Douala poverty rates are below 6% (INS, 2008).

In 2011, the under-five mortality rate was 122 per 1000 live births, while 63.6% of births were attended by a skilled health worker (WHO, 2013). Again, there are many disparities across the country. While in some regions (Littoral, West, North West, Yaoundé and Douala) more than 90% of births were attended by a skilled health worker, only 25% of births in the Far North were assisted. The leading causes of mortality and morbidity are malaria, anaemia, human immunodeficiency virus (HIV) infection and acquired immunodeficiency syndrome (AIDS), lower respiratory infections and diarrhoeal diseases (INS & ICF International, 2012). The life expectancy at birth was 50.58 years for males and 52.63 years for females in 2011.

2.1 Health system

The Cameroon health policy aims to improve the health of the population, by scaling up accessibility to quality and integrated care for the entire population, with the full participation of communities in the management and financing of health activities. It includes public, private and traditional medicine sectors.

The national health system has three levels, each of which has administrative structures, health facilities and dialogue structures relating to specific functions. The central level is made up of central services of the Ministry of Public Health and national hospitals. The intermediate level comprises regional delegations of public health and regional and related hospitals. Finally, the peripheral or operational level comprises district health services, district hospitals, subdivisional medical centres and integrated

¹ The local currency is the CFA franc. Equivalent US\$ amounts are calculated using the average exchange rate for August 2013 (US\$ 1 = CFAF 492.49).

health centres. The health centre is the first level of contact for the population, and offers a minimum package of activities (MPA).

The national territory is divided into 178 health districts; a health district is defined as a social and economic entity providing quality health care which is accessible to the public, with the full participation of beneficiaries.

The national health system was disrupted during the 1980s, a period marked by economic crisis. This situation led to, among other things, a decrease in civil servants' salaries, a hiring freeze, the closure of some training courses for health personnel, a lack of motivation among staff and various professional conflicts.

2.2 Health workforce

There was a sharp decline in the number of public sector employees in the health sector between 1992 and 2001, from 18 247 to 11 016 workers (see section 5.5). This fall of 39.6% was caused by the structural adjustment plans implemented in response to the economic crisis. The increase in numbers observed from 2003 reflects the resumption of recruitment, first on HIPC and C2D funds (in 2002, 2004 and 2007), then directly to the public sector and, finally, through integration into the public sector of staff on temporary contracts in public health facilities. Throughout the whole period, only graduated medical staff from the Faculty of Medicine and Biomedical Sciences (FMBS) of the University of Yaoundé I were recruited to the public service without interruption.

At the same time, population growth continued – from 12 million inhabitants in 1992 to 19.4 million in 2010, an increase of (61.6%) – increasing the demand for health care, within a system constrained by a shortage of human resources and an absence of investment required to improve the technical infrastructure of the public health facilities.

In 2012 the health sector had 71 medico-sanitary personnel training schools, with 41 public and 30 private institutions. Access to these schools is through a competitive national examination. Both public and private training schools are licensed by the Government and graduates may be recruited by both the private and the public sectors, since the training is regulated by the Government and all diplomas are issued by the Ministry of Public Health. The 71 training schools trained 22 687 health workers between 2000 and 2011. According to the 2011 census of national health system personnel, there were then 38 207 health workers, with 25 183 (66%) in the public sector and 13 024 in the private sector (Ministry of Public Health, 2011a) for a population of nearly 20 million inhabitants.

A better-functioning labour market is important if health outcomes are to be improved. Empirical literature suggests that a key factor in an effective and high-performing health system is the productivity of the health workers (Scheffler et al., 2012). Thus, governments, as well as the private sector, need to understand how to measure productivity and how to reward it accordingly.

3. The health labour market framework

This section summarizes the dynamics of the health labour market, as put forward by Scheffler et al. (2012).

An assessment of the health labour market needs to study both the demand and the supply side, in order to determine shortages (or eventually surpluses) of health workers.

The supply of health workers is determined by the number of qualified health workers (doctors, nurses and other care providers) willing to work for a given wage rate in the health care sector. Thus, training is a key determinant of this aspect of the labour market. The number of trained health workers depends on many factors, including the number of training institutions, the length of the training, the educational level, the cost of training, the individual interest in working in the area and the expected probability of getting a job after training.

The demand for health workers, which is linked to the demand for health care, is measured in terms of the hiring of health workers by public and private institutions. Each of these institutions competes with different wage rates, budgets, provider payment practices, labour regulations and rules that determine hiring and wage decisions.

In general, the higher the wage, the larger the number of available health workers willing to work for the health sector. But additional considerations, including working conditions, safety and career opportunities, will also influence the decision to work in the health sector or in another sector or even in another country.

The interaction between the supply and demand for health workers determines the wages and other compensation, the number of health workers employed, the number of hours they work, their geographical distribution and their employment settings.

4. Data

A preliminary list of indicators for the study was taken from Scheffler et al. (2012). The indicators included the number of health workers, the hours they work (by health occupation, sector, sex, location, etc.), wages paid (by government, private sector, etc.), other non-wage compensation and vacancy data.

We used data from a census of workers in the health sector, carried out by the Ministry of Public Health, in combination with information collected from some of the leading organizations in the private non-profit sector. The last census – the largest and most reliable in terms of quantity and quality of information – was in 2011; thus, 2011 was chosen as the reference year for the present study.

The supply of health workers consists of all persons, health professionals or not, who are willing to serve in the health sector. Health professionals include all graduates of schools of medicine, nursing and paramedical professions in the country since their creation, plus the balance of migration flow of health personnel between Cameroon and the rest of the world. However, while the number of graduates could

be obtained from the training facilities, it was not possible to obtain data on migration flows from administrative sources. Some information on migration was therefore taken from the available literature.

In addition, to obtain data on non-professionals who are willing to work in the health sector, together with information on their salaries, working time, etc., specific surveys would be needed. Therefore, this study is limited to the presentation of only a few components of the supply side of health care services.

In estimating the human resources for health gap, we used a study estimating personnel needs in public health facilities at the health district level, conducted by the Ministry of Public Health in 2011 (Ministry of Public Health, 2011b).

It was not possible to collect consistent updated time-series data for the analysis of the dynamics of the health labour market in Cameroon.

5. Health labour market analysis

5.1. Production

In Cameroon there are medical and biomedical training schools and medico-sanitary personnel training schools.

There are 71 educational institutions approved by the Ministry of Public Health for the training of medico-sanitary personnel (see section 2). Students are classified into different grades on the basis of the entrance examination, each grade being subdivided into majors (see Annex 1). The different grades at graduation are: nursing assistant, licensed nurse (or assistant nurse), state registered nurse, medico-sanitary technical agent and medico-sanitary technician.

Cameroon currently has ten faculties of medicine for training doctors, dentists and pharmacists, four public and six private (see Annex 1). Medical training is provided by 374 permanent teachers, 170 of whom are in the Faculty of Medicine and Biomedical Sciences (FMBS) of the University of Yaoundé. FMBS had one teacher for every 12 students in 2011. More than 80% of training programmes meet LMD standards.

From 1990 to 2009, 8453 health workers from a total of 30 338 graduates from training schools were recruited in the public sector (27.9%). It is difficult to assess recruitment of trained health workers in the private sector because of a lack of available data.

In public medical schools, the annual tuition fee varies from 50 000 CFAF (US\$ 102) for general practitioner training to 1 000 000 CFAF (US\$ 2030) for a specialist education. In the private medical schools that are training general practitioners, the lowest annual tuition fees are in the order of 900 000 CFAF (US\$ 1827).

The annual tuition fee for medico-sanitary workers ranges from about 150 000 CFAF (US\$ 294) to 250 000 CFAF (US\$ 500) for national candidates (residents) and 266 000 CFAF (US\$ 532) to 500 000 CFAF (US\$ 1000) for foreigners (nonresidents) (Table 1).

Table 1. Direct cost of tuition in public medico-sanitary schools, 2011

Major	Cost of tuition (US\$)	
	Residents	Nonresidents
State registered nurse (mental health, reproductive health, anaesthesia, ophthalmology)	500	1000
State registered nurse, medico-sanitary technician (sanitary engineering, kinesitherapy, dentistry, pharmacy, radiology)	320	800
Nursing assistant	294	532
Medico-sanitary technical agent, medical analysis major	400	666

Source: Ministry of Public Health, 2012a.

In the private sector, the fees differ from one institution to another, but in principle should not exceed 300 000 CFAF (US\$ 609) per year. However, in practice the fees vary between 300 000 CFAF (US\$ 609) and 1 000 000 CFAF (US\$ 2030). Additional costs are in many cases due to ancillary services to pay at the training institution.

The public sector is the largest employer. Graduates of training schools are not all automatically recruited, and the enthusiasm of school leavers and young graduates to compete in this sector is related to their chances of being hired at the end of their training.

In 2011, nearly 13 000 candidates applied to medico-sanitary training schools, and 45.5% of them passed the entrance examination. The nursing majors (nursing assistant and state registered nurse) accounted for approximately 79% of applicants and 75% of those admitted. This number of applicants to a “nursing” major seems to be low compared with the number of unemployed graduates who have the level of education required for these courses.

Table 2. Success rates in the entrance examination for medico-sanitary schools, Cameroon, 2011

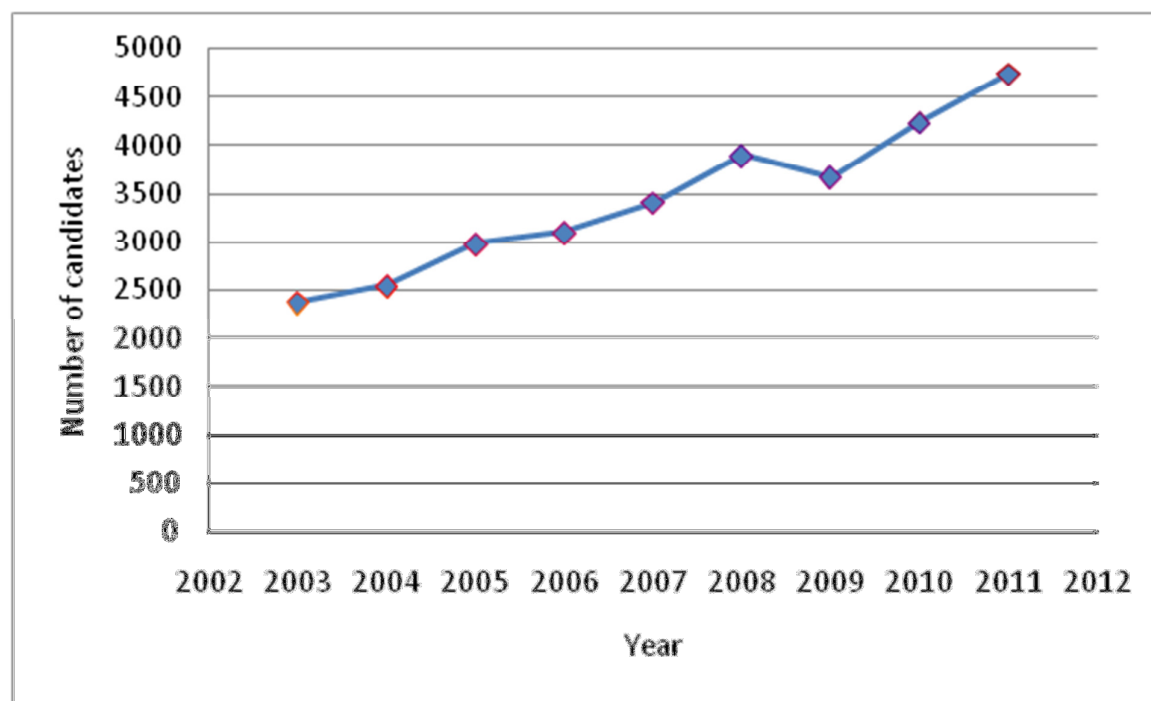
Diploma	No. of applicants	No. admitted	Success rate (%)
Nursing assistant	5394	2201	40.8
Medico-sanitary technical agent	1605	853	53.1
State registered nurse	4840	2213	45.7
Medico-sanitary technician	1100	623	56.6
Total	12 939	5890	45.5

Source: Ministry of Public Health, 2012a.

From 2003 to 2011, there was an almost constant upward trend in the number of applicants to study medicine at the FMBS, the main faculty for training doctors until 2011 (Fig. 1). Admission rates are low (4% in 2011). Possible reasons for this are the low capacity of the infrastructure, the attractiveness of the medical profession, and the low absorption rate of doctors in the public service. The numbers of

candidates reflect only those who were allowed to compete; candidates with a science high-school diploma or equivalent, and who were enrolled at the university but did not fulfill the age, or applicants violating the non-repetition and admission to the first session criteria, were not allowed to compete.

Figure 1. Number of candidates in the FMBS entrance competition, 2003 to 2011



Source: Faculty of Medical and Biomedical Science (FMBS), 2012.

All the graduates of the FMBS are automatically integrated into the public service. Each year, the public sector absorbs fewer than 100 doctors trained in the FMBS, although there is a slight overall upward trend (Table 3).

Table 3. Public sector recruitment of medical officers (generalists) trained in the FMBS

	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
No. of medical officers (generalists) recruited	73	70	71	86	77	75	93	96	86	96

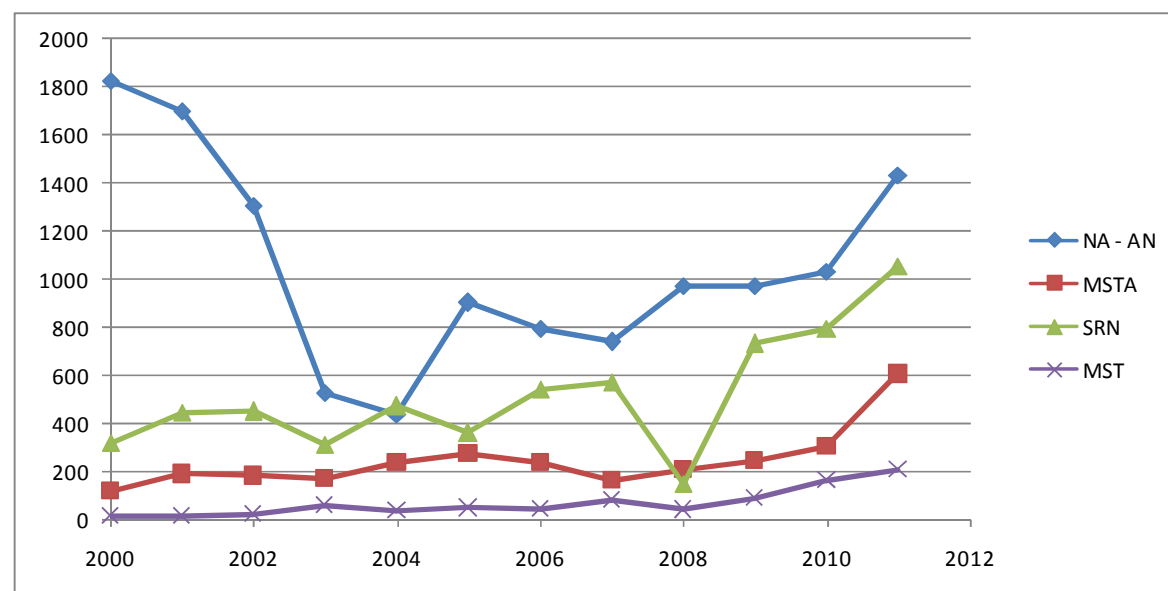
Source: Ministry of Public Health, 2012a.

The total number of medical officers trained in the FMBS is estimated to be 2320, out of a total of 2470 in the country as a whole (Nko'o Amvene, 2012). In addition to the FMBS, there are nine other faculties of medicine, three public and six private. Some of them produced their first graduates in 2010; the others have not yet produced graduates.

Projections suggest that nearly 800 trained physicians will graduate each year from 2015, which will make available just over 4000 doctors by 2020; approximately 50% of these will come from private training institutions.

The available data indicate that 22 687 medico-sanitary workers were trained during 2000–2011; 55.8% were nursing assistants and assistant nurses. The annual number of graduates in this group decreased from 2000 to 2003 as a result of the closure of a number of assistant nurse majors (general practitioners and midwives). State registered nurses account for 27.4% of the workforce trained over the period. These two groups of workers constitute the bulk of staff working in integrated health centres in Cameroon (Fig.2).

Figure 2. Number of graduates from health training schools, Cameroon, 2000–2011



NA – AN: nursing assistant – assistant nurse

MSTA: medico-sanitary technical agent (assistant laboratory technician)

SRN: state registered nurse

MST: medico-sanitary technician (laboratory technician)

Source: Ministry of Public Health, 2012a.

Since 2000, the Government has granted subsidies to private sector investments in health training institutions, in an attempt to increase the production of health workers. The subsidies account for about 1% of the financial resources of the private sector. The main effect was a 44.7% increase in the number of trained medico-sanitary workers, from 2285 to 3307 from 2000 to 2011.

The most important remaining problem in medical training is its quality. Early this year the Cameroon Medical Council assessed the medical training institutions, and concluded that, apart from the four public faculties of medicine, only two private institutions met the minimum standards of medical training.

The Government still allows automatic recruitment of graduates from public faculties. Considering the emphasis currently being put on the quality of medical training, and in order to reduce the workforce shortage as part of efforts to reach the MDGs, more efforts should be made by the public authorities to hire graduates from private institutions; this will also boost private investment in the health sector.

5.2. Health workers by category

Of 38 207 health workers in the national health system in 2011, nearly half were nurses or equivalent. The next biggest groups were the supportive staff (17.5%) and the paramedical staff (11.8%). Only 4.8% of staff were medical doctors (Table 4).

Table 4. Number of health workers by type and sector

Type	Public	Private			Total	%
		Non-profit	For profit	Total		
Medical officer	1471	162	209	371	1842	4.8
Generalist	1132	132	156	288	1420	3.7
Specialist	339	30	53	83	422	1.1
Dental surgeon	39	9	10	19	58	0.2
Pharmacist	55	22	85	107	162	0.4
Nurse	13 084	3149	2721	5870	18 954	49.6
Paramedical staff	3127	751	648	1399	4526	11.8
Pharmacy attendant	900	162	116	278	1178	3.1
Other health professional	1164	969	647	1616	2780	7.3
Social assistant	83	14	8	22	105	0.3
Administrative staff	1250	288	186	474	1724	4.5
Support staff	3866	1737	1070	2807	6673	17.5
Other	144	3	58	61	205	0.5
Total	25 183	7266	5758	13 024	38 207	100.0

Source: Ministry of Public Health, 2011a.

Nearly two out of every three health workers (65.9%) are employed in the public sector. Of the 13 024 health workers in the private sector, 55.8% (7266) are in non-profit private facilities (faith-based associations and nongovernmental organizations (NGOs)) (Table 4). Nurses represent nearly 52% of the workforce in the public sector; 69% of all nurses work in the public sector.

5.3. Health workforce by age and sex

In 2011, 36.1% of the national health workforce was aged between 31 and 40 years; around two-thirds were between 31 and 50 years old. In the age group up to 50 years, women were in the majority, while among those between 51 and 65 years, men were more numerous (Table 5). In other words, the new entrants in the health professions are predominantly women.

Table 5. Distribution of health workers by age and sex

Age group	Female		Male		Total	
	No.	%	No.	%	No.	%
< 20	191	0.9	91	0.5	282	0.7
21–30	4949	23.0	2801	16.7	7750	20.3
31–40	8300	38.6	5491	32.8	13 791	36.1
41–50	5899	27.5	5334	31.9	11 233	29.4
51–55	1447	6.7	1810	10.8	3257	8.5
56–60	478	2.2	678	4.1	1156	3.0
61–65	77	0.4	248	1.5	325	0.9
≥ 65	41	0.2	175	1.0	216	0.6
Not specified	102	0.5	95	0.6	197	0.5
Total	21 484	100.0	16 723	100.0	38 207	100.0

Source: Ministry of Public Health, 2011a.

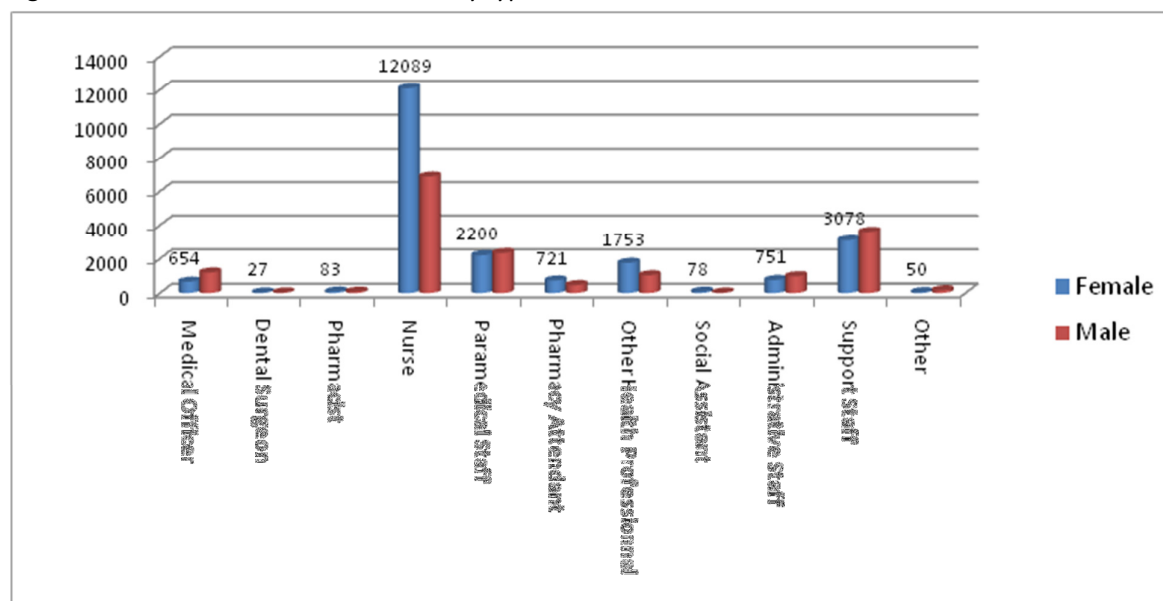
Women account for more than 56% of all staff in the national health system – 58% in the private sector and just over 55% in the public sector (Table 6). As shown in Figure 3, this preponderance of female workers is the result of the high proportion of “nurses and assimilated” in the system.

Table 6. Distribution of health workers by sex and sector

Sex	Public sector		Private sector		Total	
	No.	%	No.	%	No.	%
Male	11 234	44.6	5489	42.1	16 723	43.8
Female	13 949	55.4	7535	57.9	21 484	56.2
Total	25 183	100.0	13 024	100.0	38 207	100.0

Source: Ministry of Public Health, 2011a.

Figure 3. Distribution of health workers by type and sex



Source: Ministry of Public Health, 2011a.

The preponderance of women in the health workforce presents challenges for human resources management, especially regarding the reconciliation of maternity constraints and administrative provisions, such as family reunification, with the requirements for service performance. Measures such as task shifting and the use of temporary staff should be explored in an attempt to respond to the challenges.

5.4. Geographical distribution of the health workforce

Cameroon is currently decentralizing services, with the aim of allocating more and more resources (financial, material and human) to local authorities. It is thus appropriate to analyse the geographical distribution of the health workforce in relation to that of the population to be served.

There is strong evidence of workforce imbalances in the country. The Centre and Littoral regions, with the largest hospitals in Cameroon, account for a total of 42.5% of the health workforce (24.3% and 18.2%, respectively). In terms of population, the Centre is at the top of the ranking, with 18.3% of the estimated population in 2012. However, Littoral is only in third position, with 14.9% of the population. The Far North is the second most populous region, with 18.0%, but has only 9.8% of health workers; the North, which has 10.9% of the population, has only 4.2% of the health workforce.

With 3.2 health workers per 1000 population, Yaoundé is the most well served city in terms of health workers. Yaoundé is the country's administrative capital, and the two largest hospitals of Cameroon are located there. Yaoundé and Douala (which has 2.3 health workers per 1000 population) have the lowest poverty rates (5.9% and 5.7%, respectively, in 2007) and among the best maternal health indicators (more than 9 out of 10 births took place in a health facility in 2011).

The most striking shortages of health workers are in the Far North, the North and Adamawa. These are the poorest regions in the country: the poverty rate in 2007 was 65.9% in the Far North, 63.7% in the North and 52.9% in Adamawa. They also have the lowest densities of human resources for health: 1.1 health workers per 1000 population in Adamawa and 1 per 1000 population or fewer in the two other regions.

In addition, children in the northern regions are the most affected by diarrhoeal diseases, one of the leading causes of death in children (Table 7). The estimated prevalence of diarrhoeal diseases in the North and Far North is 35%, and in Adamawa 20%. These three regions are the most affected, followed by the East (18%) and the South (17%). The regions where children are less affected by diarrhoeal diseases are the Littoral (8%), West (10%) and South West (10%).

Table 7. Selected characteristics of the regions

Region	No. of health workers per 1000 population (2012)	Poverty rate (2007)	% of poor people (2007)	% of births attended by trained health worker (2011)	% of births occurring in a health facility (2011)	% of children with diarrhoea treated in a health facility or by a health worker (2011)
Yaoundé	3.2	5.9	1.4	92.9	90.9	33.1
Douala	2.3	5.5	1.4	98.8	97.8	25.6
Adamawa	1.1	52.9	6.9	47.4	45.8	31.1
Centre ¹	1.5	41.2	7.9	78.5	71.8	21.7
East	1.8	50.4	5.9	48.9	46.2	32.5
Far North	1.0	65.9	29.9	25.1	22.7	15.7
Littoral ²	2.3	30.8	2.7	94.2	92.1	37.9
North	0.7	63.7	15.7	32.9	30.2	15.3
North West	2.1	51.0	13.0	93.6	93.7	39.9
South	1.8	29.3	2.4	82.2	77.2	37.1
South West	2.6	27.5	5.2	80.1	78.1	31.3
West	2.8	28.9	7.7	95.8	93.9	36.0
Total	1.9	39.9	100.0	63.6	61.2	22.8

Source: Ministry of Public Health, 2011a; BUCREP, 2010; INS, 2008, 2011.

¹ Excluding Yaoundé.

² Excluding Douala.

In the North, Far North, Adamawa and East regions, more than 61% of births took place at home in 2006 (INS, 2006). A high percentage of deliveries are assisted by traditional birth attendants or relatives and friends in rural areas. This is reflected by the low share of births assisted by trained health workers in the Far North (25.1%) and the North (32.9%) (see Table 7). Clearly, more health workers are needed in these regions. Other regions e.g. Douala (98.8%), West (95.8%), Littoral (94.2%), North West (93.6%) and Yaoundé (92.9%) show very high birth attendance by trained health workers.

There is thus evidence of inequitable allocation of human resources in the system and its consequences. During the past decade, a number of policies have been implemented to address migration and maldistribution of human resources. One example is the emergency plan for upgrading quantitative and qualitative health workforce. Under this plan, in 2009, 2967 personnel from foreign and national private and public training schools were recruited into the public service, following a direct competitive examination. In addition, a recruitment process was launched for some 3000 employees working on short-term contracts. These programmes were funded by development partners under the Contract Debt Relief and Development (C2D) and the Heavily Indebted Poor Countries Initiative (HIPC); once they came to an end, the staff were recruited into the public service, at which point they were free to move to a different area. In 2012, a new programme was launched, supported by the C2D funding, to encourage employment in difficult areas.

5.5. Health workforce by sector

Table 8 shows the change in the size of the health workforce in the public sector since 1992. There was a sharp decline of 39.6% between 1992 and 2001, as a result of structural adjustment programmes implemented in response to the economic crisis. Investment in the health infrastructure also stagnated. At the same time, the population continued to grow, resulting in increased demand for health care.

Table 8. Employment in the public sector, 1992–2011

Year	1992	1993	1997	2001	2003	2005	2007	2009	2010	2011
No. of health workers in public sector	18 247	16 802	14 292	11 016	11 972	11 528	14 154	15 720	19 709	25 183

Source: Ministry of Public Health, 2012a.

The upward trend in numbers from 2003 reflects the resumption of recruitment, first using HIPC and C2D funds (in 2002, 2004 and 2007), then directly to the public service, and finally through integration into the public service of temporary personnel. Only graduates from the Faculty of Medicine and Biomedical Sciences were automatically recruited throughout this period.

Most personnel in the national health system (88.5%) are working in health care facilities. The equivalent proportion in the private sector is almost 94%. The next biggest group is in administrative services; these coordination and organization activities are almost exclusively carried out in the public sector (2908 administrative workers out of a total of 2939 (98.9%)).

Table 9. Distribution of health workers by type of facility and sector

Type of facility	Public		Private		Total	
	No.	%	No.	%	No.	%
Care delivery	21 612	85.8	12 211	93.8	33 823	88.5
Administrative	2908	11.5	31	0.2	2939	7.7
Drugstore and assimilated	155	0.6	594	4.6	749	2.0
Training or research	424	1.7	118	0.9	542	1.4
Other	84	0.3	70	0.5	154	0.4
Total	25 183	100.0	13 024	100.0	38 207	100.0

Source: Ministry of Public Health, 2011a.

- **Dual practice**

Treatment is still significantly better in the private sector than in the public sector. Salaries in the public sector are relatively low, and many public workers take part-time jobs in the private sector to supplement their income. This leads to high absenteeism in the public sector, and consequently reduced productivity and performance.

5.6. Migration

The migration of health professionals has been a cause for concern to health authorities in recent years. Abena Obama et al. (2003) found that 49.3% of the health professionals in Cameroon intended to emigrate.

During 1991–2000, 173 doctors (mostly men, aged between 20 and 40 years), 50 dentists, 155 nurses (mostly specialist nurses in, for example, paediatrics, intensive care and anaesthetics), 50 midwives and nine pharmacists (aged between 20 and 40 years) emigrated. The main destinations of those who emigrate are the United Kingdom, the United States of America, France and Belgium (Awases et al., 2004).

Since 1994, the Government has tried to address migration through the payment of a productivity allowance amounting to 10% of the financial resources generated by the facility.

5.7. Wages

The structure of wages in the health sector varies considerably in Cameroon. Workers in the private sector are far better off than their counterparts in public institutions. In fact, public sector workers are on the same salary scale as all other employees in the public services. As an illustration, a doctor in the para-public sector earns at least 2.1 times more than a doctor in the public service with the same qualifications, and over 3.5 times more than a contract-holder with the same technical profile (Table 10). Within the public sector, contract-holders are less well paid than civil servants with equal skills and equal performance.

Table 10. Monthly wages paid to health workers, by sector^a

Type of health worker		Public sector				Para-public sector		Private			
		Civil servant		Contract-holder		NSIF ^b		For profit		Non-profit	
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
Generalist medical officer		483	759	305	453	1135	1696	1137	1732	606	1060
Specialist medical officer		466	692	275	441	976	1583	817	1444	520	725
Dental surgeon		457	639	264	401	937	1447	817	1010	-	-
Pharmacist		466	692	275	421	976	1583	817	1085	-	-
Nursing officer		374	626	257	419	791	1021	462	562	437	765
State registered nurse		344	528	234	384	600	851	387	528	417	745
Nursing assistant – assistant nurse		179	322	122	245	367	581	240	387	278	324
Technical medico-sanitary agent		166	307	109	221	291	479	212	378	167	285
Medico-sanitary technician		344	528	212	316	589	833	315	462	417	745
Sanitary engineering technician		337	491	198	288	547	811	315	462	-	-
Biomedical technician		337	491	198	288	567	838	315	462	417	745
Biomedical works engineer		378	628	271	403	733	1,167	771	1034	-	-
Biomedical engineer		403	680	285	419	987	1,298	-	-	-	-
Public health administrator		400	642	-	-	749	1,197	-	-	-	-
Health auxiliary		-	-	-	-	-	-	119	163	76	92

^a Wages are expressed in US\$, based on an exchange rate of US\$1 = 500 CFAF.

^b National Security and Insurance Fund

Sources: Public service salary scales, NSIF, Catholic Health Services in Cameroon (CHOC), Council of Protestant Churches in Cameroon (CPCC), Cameroon Ad Lucem Foundation (CALF).

In an attempt to improve both the productivity and the salaries of health workers, performance-based financing (PBF) has been implemented by the Catholic Organization for Relief and Development Aid (CORDAID) in the East Region since 2006. Since 2011, the approach has extended to a number of pilot sites by the World Bank, in order to increase the quality and quantity of services, through the purchase of care “on the basis of a contract between the health facilities and the agency”.

The mid-term review of PBF in the East, North West, South West and Littoral regions noted the following effects:

- improved quality of care and service through the monitoring of predefined performance indicators;
- renewed motivation among health professionals;
- better organization of work (exploitation of rigorous sheets, clear definition of tasks, etc.);
- increased attendance and the utilization of care and health services (up to threefold)
- fewer unregulated fees; use of official rates for services and medicines;
- involvement of health workers in decision-making on the management of the facility.

The impact study will evaluate the benefits of this approach and, if appropriate, develop proposals to scale up PBF nationwide.

5.8. Hours worked by health workers

In the public sector, the current regulations specify a 40-hour working week; however, in practice, the actual working hours depend on the type of structure and area of residence.

Table 11. Hours worked per week, by type of health worker and by sector

Type of health worker	Public sector	Private sector	
		Non-profit	For profit
Generalist medical officer	40	40–45	40–45
Specialist medical officer	40	30–45	30–45
Dental surgeon	40	40–45	40–45
Pharmacist	40	40–45	40–45
Nurse	40–56	30–48	40–72
Paramedical staff	30–50	30–48	40–60
Other health professional	40	40–45	40–60

Sources: NSIF, CHOC, CPCC, CALF and our survey 2011.

Working hours in secondary and tertiary level health facilities (general and central hospitals) are generally lower than those in integrated health centres (IHCs) and subdivisional medical centres (SMCs). Moreover, in rural areas, which tend to have more IHCs and SMCs, there are often fewer patients than in urban areas the facilities have less staff, which means that the staff have to perform beyond the regulatory standard; in some cases, nurses work 56 hours a week.

In the private sector, working hours are on average somewhat higher, especially in for-profit centres, where nurses may be working over 70 hours per week (Table 11).

5.9. Health workers shortages and surpluses

The most common measure for identifying whether there are economic shortages or surpluses of health workers in a health labour market is the vacancy rate, which is defined as the ratio of the number of unfilled vacancies to the number of funded health care posts. This allows the gap between the demand and the supply of health workers to be identified. Unfortunately, Cameroon has no data on the vacancy rate.

- **Needs-based shortage**

Data are, however, available to estimate the deficiency of the health system to cover the needs of the population, the needs-based shortage. The needs-based shortage, measures the gap between the available health workforce and the health workforce required to meet the needs of the population. There are several models for estimating the need for human resources for health needs:

- the ratio of personal health / populations;
- use of epidemiological and demographic forecasts;
- projected use of health services;
- use of health services' specific objectives.

The Ministry of Public Health in Cameroon focused on identifying needs based on the workload corresponding to the achievement of the health services' specific objectives, i.e. activities related to MDG4 (reduction of child mortality), MDG5 (reduction of maternal mortality) and MDG6 (reduction of malaria, HIV/AIDS and tuberculosis).

The workload was calculated by the Workload Indicators of Staffing Needs (WISN) method. Developed by WHO, this method is based on measuring the individual workload of health personnel and on the standard time required to complete each component of the overall workload.

Based on workers' standards (by category of health facility and area of residence), which were revised in 2011, and taking into account the health district level only, overall staffing needs were obtained for integrated health centres, subdivisional medical centres and district hospitals (Table 12).

Table 12. Staffing needs in IHCs, SMCs and district hospitals, 2011

Type of health worker	Staff required		Staff available		Shortage	
	No	%	No.	%	No.	%
Generalist medical officers	727	2.5	382	2.9	345	2.1
Specialist medical officers	468	1.6	34	0.3	434	2.7
Dental surgeons	156	0.5	17	0.1	139	0.9
Nurses	14 025	47.8	8069	61.0	5956	37.0
Paramedical staff	5188	17.7	1704	12.9	3484	21.6
Pharmacy attendants	1801	6.1	827	6.3	974	6.1
Social assistants	156	0.5	39	0.3	117	0.7
Administrative staff	805	2.7	180	1.4	625	3.9
Support staff	5992	20.4	1972	14.9	4020	25.0
Total	29 318	100.0	13 224	100.0	16 094	100.0

Source: Ministry of Public Health, 2011b.

In 2011, there was a national shortage of 16 094 workers at district level in the public sector, representing the difference between the estimate of needs based on updated workload standards 29 318 and available staff (13 224) (Ministry of Public Health, 2011b). In other words, the public sector only had 45% of the health workforce required to meet the health needs of the population.

Of the 16 094 extra health workers needed, 37% are nurses (assistant nurses, nursing assistants, state registered nurses and midwives), 21.6% paramedical staff (medico-sanitary technical agents and medico-sanitary technicians) and 4.8% medical doctors (Table 12).

Scheffler et al. (2008) predicted that Cameroon would experience a needs-based shortage of 9625 physicians and a demand-based shortage of 717 physicians in 2015.

6. Main findings and discussion

The most important remaining problem in medical training is its quality. Early this year the Cameroon Medical Council assessed the medical training institutions, and concluded that, apart from the four public faculties of medicine, only two private institutions met the minimum standards of medical training.

The Government still allows automatic recruitment of graduates from public faculties. Considering the emphasis currently being put on the quality of medical training, and in order to reduce the workforce shortage as part of efforts to reach the MDGs, more efforts should be made by the public authorities to hire graduates from private institutions; this will also boost private investment in the health sector.

The total number of medical officers trained in the FMBS is estimated to be 2320, out of a total of 2470 in the country as a whole. The rest were trained in the University of Mountains (UDM). All physicians trained in the public university FMBS are hired by the Government. In spite of the low success rate in the entrance examination (about 4% in public faculties), there is a gradual increase in the number of applicants to medical school every year.

A total of 22 687 medico-sanitary workers were trained during 2000–2011; 55.8% were nursing assistants and assistant nurses. State registered nurses were the next biggest group, at 27.4% of all those trained. There is a downward trend in the number of graduates of this subgroup since 2003 due to the closure of some training fields and the low employment rate of health workers.

Most personnel in the national health system (88.5%) are allocated to health care facilities. Of 38 207 health workers in the national health system in 2011, nearly half were nurses; supportive staff accounted for 17.5% of workers, and paramedical staff for 11.8%.

Nearly two-thirds of health workers (65.9%) are employed in the public sector. Of the 13 024 health workers in the private sector, 7266 (55.8%) are in non-profit facilities (faith-based associations and NGOs). Nurses account for nearly 52% of the workforce in the public sector; and 69 % of all nurses work in the public sector. Measures such as task shifting and the use of temporary staff should be explored in

an attempt to overcome this constraint in the short term. Fulton et al. (2011) have suggested a series of questions that can help with task shifting in a low-income country.

There is strong evidence of workforce imbalances in the country. The Centre and Littoral regions, with the largest hospitals in Cameroon, account for a total of 42.5% of the health workforce (24.3% and 18.2%, respectively). In terms of population, the Centre is at the top of the ranking, with 18.3% of the estimated population in 2012. However, Littoral is only in third position, with 14.9% of the population. The Far North is the second most populous region, with 18.0%, but has only 9.8% of health workers; the North, which has 10.9% of the population, has only 4.2% of the health workforce.

The consequences of this inequitable distribution of human resources are dramatic. For example, Adamawa, the Far North and the North, which are among the poorest regions of Cameroon, also have the highest prevalence of diarrhoeal diseases, especially in children. Around half or more of deliveries take place at home and are not assisted by a health worker. Cameroon is currently engaged in a decentralization policy, which aims to allocate more and more resources (financial, material and human) to local authorities. Special efforts should be made by the public authorities to reduce the discrepancies in the geographical distribution of the health workforce.

Working hours in the private sector are on average somewhat higher than in the public sector, especially in for-profit facilities, where some nurses may be working over 70 hours per week. Since incomes in the public sector tend to be relatively low, a large proportion of workers hold a second part-time job in the private sector; the main consequence is increased absenteeism in the public sector. However, there have so far been no studies to investigate this phenomenon.

There are inequalities in the wage structure. All wages in the public sector follow the public services scale. A doctor in the para-public sector, whether general practitioner or specialist, earns at least 2.1 times more than a doctor in the public service with the same qualifications, and over 3.5 times more than a contract-holder with the same technical profile. Within the public sector, contract-holders are less well paid than civil servants with equal skills and equal performance. In general, as is the case in many developing countries, earnings are low despite long work hours (Fields, 2010). All the above factors are likely to reduce the productivity and performance of public health facilities.

Many Cameroonian health workers either migrate or take up a secondary activity in order to face the “African family pressure”. For those who work in the public sector, this secondary activity will often be in a private clinic (dual practice), again reducing the productivity and performance of public health facilities. Attempts have been made to tackle emigration, including by improving opportunities for professional advancement, so as to reduce the number of professionals travelling abroad to further their studies (Amani, 2010).

However, there are some limitations to the results reported in this paper. The main weakness was the dearth of data. Accurate time-series data for analysis of labour market dynamics were unavailable. Data on migration were not up to date, there were no data on the informal economy or the vacancy rate, and there was very little information on dual practice. Also, factors affecting the attractiveness of the medical field have not been studied. Job security may be one of the major factors, but that remains to

be proven. Thus, for a better understanding of the labour market in the health sector, further studies should be conducted.

7. Conclusions

In considering the supply of health workers, an important element is the attractiveness of the health professions. Fewer than half of applicants to medico-sanitary training schools pass the entrance examination, while only around 4% of applicants to medical school are successful. Since the public sector is the largest employer, the enthusiasm of school-leavers and young graduates to compete in this sector is related to their perception of the probability of being hired at the end of the training.

There is an uneven geographical distribution of the health workforce, as well as an overall shortage of health workers. The actual working hours of health workers depend on the type of structure and area of residence. In rural areas, there are often fewer people than in urban areas, which means that the existing staff has to perform well beyond the regulatory standard. Working hours tend to be higher in the private sector. In 2011, the overall shortage of staff in the health sector, based on the level required to achieve the MDGs, was estimated at 16 094, of whom 37% were nurses.

The wage structure is quite variable. Workers in the para-public sector are far better off than their counterparts who work in the public administration or public health facilities. Within the public sector, contract-holders are less well paid than civil servants with equal skills and equal performance. Treatment of patients is still significantly better in the private sector than in the public sector.

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Annex

1. Majors associated with different grades of medico-sanitary personnel

Nursing assistant

- General practitioner
- Laboratory option (closed in 2002)

Licensed nurse (assistant nurse)

- General practitioner (closed in 2003)
- Midwife (closed in 2004)

State registered nurse

- General practitioner
- Reproductive health
- Anaesthesia resuscitation
- Ophthalmology
- Mental health
- Midwife (started in 2011)

Medico-sanitary technician

- Medical analysis
- Health engineering
- Kinesitherapy
- Pharmacy
- Dentistry
- Psycho-traction and relaxation
- Medical Imaging
- Optics and refraction (started in 2011).

2. Faculties of medicine that train doctors, dentists and pharmacists

- Faculty of Medicine and Biomedical Sciences of the University of Yaoundé 1 (public)
- Faculty of Medicine and Pharmaceutical Sciences of the University of Douala (public)
- Faculty of Health Sciences of the University of Buea (public)
- Faculty of Medicine and Health Sciences of the University of Bamenda (public)
- Faculty of Medical Sciences of University of Mountains of Bangangté (private)
- Higher Institute of Medical Technology of Nkolondom (private)
- Faculty of Medical, and Medico-sanitary Pharmaceutical Sciences of Ecuador University (private)
- Faculty of Medicine of Mbo plain (private)
- Faculty of Medicine of the Edwin Cozzens Protestant University of Elat (private)
- Higher Institute of Health Professions (private).